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I. ABSTRACT

This paper implements and analyzes a two-dimensional nine-point (9-point) stencil for heat diffusion, first as a serial baseline and then as a shared-memory parallel program using POSIX threads (pthreads). The serial version computes a non-weighted mean of each cell and its eight neighbors per timestep, writing the full time evolution to a stacked binary format for visualization and verification. We extend this implementation to pthreads via row/block domain decomposition with barrier synchronization at iteration boundaries, preserving correctness while exposing thread-level concurrency consistent with Pacheco’s methodology for shared-memory design [?]. We define speedup, efficiency, and iso-efficiency for quantitative evaluation on multicore CPUs and outline a testbed on SDSC’s Expanse system. Results establish a reproducible serial reference and a clear path to scalable parallel execution, emphasizing decomposition, synchronization, and experimental rigor.

Index Terms—9-point stencil, high-performance computing (HPC), parallel programming, POSIX threads (pthreads), shared-memory systems, scalability, speedup, efficiency, iso-efficiency, synchronization, load balancing, data parallelism, Expanse cluster, performance analysis, diffusion simulation.

II. INTRODUCTION

Stencil computations are a foundational pattern in scientific computing and a practical vehicle for introducing shared-memory parallel programming [?]. In a stencil, each grid point is updated from a fixed neighborhood, producing a highly regular iteration space with well-defined data dependencies. We employ a two-dimensional 9-point stencil (center plus eight neighbors) to model heat diffusion through a non-weighted local average per timestep.

Our goals are threefold: (i) implement a *serial* 9-point stencil that is correct, profiled, and reproducible; (ii) extend the kernel to a *pthreads* version through domain decomposition and barrier synchronization; and (iii) measure and interpret *speedup*, *efficiency*, and *iso-efficiency* on multicore CPUs, including SDSC Expanse.

A. Problem and Data Model

Let $A^{(t)} \in \mathbb{R}^{N \times M}$ be the grid at timestep t . For interior indices (i, j) ,

$$A_{i,j}^{(t+1)} = \frac{1}{9} \sum_{p=i-1}^{i+1} \sum_{q=j-1}^{j+1} A_{p,q}^{(t)},$$

while boundary values remain fixed (Dirichlet boundaries). All I/O uses a compact binary format: a header with two 32-bit integers (`rows`, `cols`) followed by $N \times M$ doubles (row-major). The stacked output encodes the full time evolution for post-processing and visualization.

B. Contributions

- A clean serial baseline with double buffering and per-iteration frame stacking.
- A pthreads design that preserves serial semantics via domain decomposition and barrier synchronization, aligned with Pacheco’s shared-memory guidance [?].
- Definitions and methodology for speedup, efficiency, and iso-efficiency to evaluate scalability on multicore nodes, with SDSC Expanse as the target platform.

III. BACKGROUND AND RELATED WORK

Stencil computations discretize spatial operators over regular grids and are fundamental in applications such as heat diffusion, seismic imaging, and fluid flow. Their local-neighborhood structure makes them ideal for studying memory locality, synchronization, and scalability [?], [?], [?].

Pacheco highlights a disciplined methodology for building parallel programs: begin with a correct serial baseline, validate its numerical accuracy, and only then introduce concurrency through domain decomposition and synchronization primitives [?]. This serial-to-parallel evolution forms the foundation of the present study.

Datta et al. [?] modeled stencil performance and emphasized the importance of blocking, vectorization, and cache reuse on CPUs. Stengel et al. [?] employed Execution-Cache-Memory (ECM) models to identify bottlenecks, while Shalf and Olicker [?] analyzed communication and cache effects in parallel stencil codes. More recent work by Biswas and Datta [?] optimized shared-memory stencils through data privatization and thread affinity.

Unlike these optimization-focused efforts, this project adopts a methodological emphasis: establishing a validated serial 9-point stencil and then introducing a pthreads-based shared-memory parallelization. The objective is to measure strong-scaling behavior—speedup, efficiency, and iso-efficiency—on a modern multicore system.

IV. PERFORMANCE METRICS

The performance of the pthreads implementation is evaluated using four interrelated metrics: *execution time*, *speedup*, *efficiency*, and *iso-efficiency*. Together they describe both the immediate performance gains from parallelization and the longer-term scalability trends.

A. Execution Time

The wall-clock runtime using P threads is denoted T_P . The single-threaded baseline, T_1 , serves as the reference. All timings are collected using high-resolution timers defined in `timer.h` and averaged over multiple trials to minimize transient effects such as OS jitter and cache warm-up variability.

B. Speedup

Speedup measures the factor by which runtime decreases when using P threads:

$$S(P) = \frac{T_1}{T_P}. \quad (1)$$

Ideally, $S(P)$ increases linearly with P ; however, serial portions and synchronization costs limit achievable performance.

a) Amdahl's Law. Amdahl's 1967 model [?] expresses this limitation quantitatively:

$$S(P) = \frac{1}{(1-f) + \frac{f}{P}}, \quad (2)$$

where f is the parallel fraction of the program. The non-parallel portion $(1-f)$ bounds maximum speedup even if infinite threads were available.

C. Efficiency

Efficiency quantifies resource utilization:

$$E(P) = \frac{S(P)}{P} = \frac{T_1}{PT_P}. \quad (3)$$

An efficiency near one indicates excellent scaling; lower values reveal idle time, load imbalance, or synchronization overhead. To analyze losses, we define the parallel overhead

$$\Theta(P) = PT_P - T_1, \quad (4)$$

and equivalently,

$$E(P) = \frac{T_1}{T_1 + \Theta(P)}. \quad (5)$$

Reducing $\Theta(P)$ directly improves efficiency.

D. Iso-Efficiency and Scalability

While $E(P)$ describes current utilization, *iso-efficiency* characterizes how the problem size W (total work) must grow with P to maintain a target efficiency η . Formally, the iso-efficiency function satisfies

$$W = K \cdot T_o(P), \quad (6)$$

where $T_o(P)$ represents total parallel overhead and K depends on the desired efficiency level. Substituting Eq. (??) shows that, to sustain $E(P) = \eta$,

$$T_1(W) \geq \frac{\eta}{1-\eta} \Theta(P). \quad (7)$$

Hence, if overhead doubles as P doubles, the problem size must also double to maintain the same efficiency.

In practice, iso-efficiency is observed empirically by running progressively larger grids and plotting the smallest size $W^*(P)$ achieving $E(P) \geq \eta$. A flatter $W^*(P)$ curve implies better scalability. For the 9-point stencil, near-linear iso-efficiency behavior is expected up to the physical core limit of the Expanse node.

E. Summary

Speedup, efficiency, and iso-efficiency jointly reveal the strong-scaling characteristics of the pthreads implementation. $S(P)$ quantifies raw performance gain, $E(P)$ indicates how effectively resources are used, and the iso-efficiency function projects scalability as problem size grows. These analyses, grounded in Amdahl's Law [?] and elaborated by Pacheco [?], form the theoretical basis for the experimental evaluation that follows.

V. DESIGN AND IMPLEMENTATION

A. Data Format and Initialization

All matrices use a compact binary format: `[int32 rows][int32 cols][rows*cols doubles]` in row-major order. The initializer creates `initial.dat` with interior zeros and fixed left/right edges at 1.0 to establish a gradient.

B. Serial Kernel (Baseline)

The serial stencil uses two arrays A (read) and B (write) and performs T sweeps:

- 1) Update interior via the 9-point average; keep boundaries unchanged.
- 2) Swap $A \leftrightarrow B$.
- 3) Append current frame to the stack file (initial + every timestep) for validation and visualization.

Wall clock is measured with a microsecond-accuracy timer.

C. Pthreads Parallelization

The shared-memory parallel version employs POSIX threads (pthreads), a standardized threading interface defined by the IEEE [?]. Each thread is assigned a contiguous block of rows in the 2D grid, minimizing communication and ensuring that halo dependencies are restricted to the immediate neighboring thread boundaries. This partitioning model aligns closely with the structured shared-memory methodology discussed by Pacheco [?].

Barrier synchronization is introduced between iterations to maintain data consistency across thread boundaries. Each thread independently computes its assigned subregion of the grid, then participates in a global barrier to ensure that all partial results are visible before the next timestep begins. The `pthread_barrier_wait()` call provides a lightweight synchronization primitive for this step, allowing efficient parallel execution on multicore CPUs.

All experiments were executed on the San Diego Supercomputer Center’s Expanse cluster [?], which provides AMD EPYC 7763 CPU nodes with 256 GB RAM and high-bandwidth interconnects. Threads were pinned to physical cores, and first-touch memory allocation was employed to minimize NUMA effects during benchmarking.

- **Work partition:** split $i \in [1, N - 2]$ into P contiguous stripes; each thread processes $[i_{\text{start}}, i_{\text{end}}]$ and all $j \in [1, M - 2]$.
- **Barriers:** a barrier after writing B ensures halos are ready; a second barrier after a single swap (`tid==0`) ensures all threads see the new A .
- **Boundaries:** fixed boundaries are copied from A to B by the owner threads of edge rows/columns.
- **I/O:** to keep frames deterministic, `tid==0` writes the initial and each per-iteration frame to the stack.
- **Practical notes:** padding/alignment to avoid false sharing, first-touch initialization for NUMA, and pinning threads to cores for stability [?].

D. Algorithms (Pseudocode)

- Serial 9-Point Stencil:
- Parallel 9-Point Stencil (Pthreads):

E. SDSC Expanse (CPU Node) Overview

Expanse CPU nodes feature dual-socket AMD EPYC nodes with large memory, fast local storage, and high-speed interconnect. We use CPU nodes with SLURM for batch/interactive runs and pin pthreads to cores for consistent measurements.

VI. EXPERIMENTAL SETUP AND RESULTS

A. System Environment

All experiments were conducted on the San Diego Supercomputer Center’s Expanse cluster [?]. Each compute node features dual AMD EPYC 7763 processors with 64 cores per socket (128 hardware threads total) and 256 GB of DDR4 memory. Experiments used a single node to ensure a shared-memory environment consistent with the pthreads model.

Algorithm 1 9-Point Stencil Heat Diffusion (Serial)

Require: $A \in \mathbb{R}^{N \times M}$ (fixed boundaries), timesteps T
Ensure: Updated grid after T iterations; stacked output

```

0: allocate  $B$ 
0: write initial frame ( $A$ ) to stack
0: for  $t = 1$  to  $T$  do
0:   for  $i = 1$  to  $N - 2$  do
0:     for  $j = 1$  to  $M - 2$  do
0:        $B[i, j] \leftarrow \frac{1}{9} \sum_{p=i-1}^{i+1} \sum_{q=j-1}^{j+1} A[p, q]$ 
0:     end for
0:   end for
0:   copy boundaries from  $A$  to  $B$ 
0:   swap ( $A \leftrightarrow B$ )
0:   write frame ( $A$ ) to stack
0: end for
0: write final ( $A$ ) = 0

```

Algorithm 2 Parallel 9-Point Stencil (Pthreads, Row Decomposition)

Require: P threads; A with fixed boundaries; timesteps T

```

0: allocate  $B$ ; create barrier; partition  $[1, N - 2]$  into  $P$  stripes
0: Thread 0: write initial frame
0: for  $t = 1$  to  $T$  do
0:   Thread  $tid$ : for  $i \in [i_{\text{start}}, i_{\text{end}}]$ ,  $j \in [1, M - 2]$ 
0:      $B[i, j] \leftarrow \frac{1}{9} \sum_{p=i-1}^{i+1} \sum_{q=j-1}^{j+1} A[p, q]$ 
0:   copy boundaries for owned rows (endpoints by owner threads)
0:   barrier.wait() {all  $B$  ready for swap}
0:   if  $tid = 0$  then swap ( $A \leftrightarrow B$ )
0:   end if
0:   barrier.wait() {all see new  $A$ }
0:   if  $tid = 0$  then write frame ( $A$ )
0:   end if
0: end for
0: if  $tid = 0$  then write final ( $A$ )
0: end if=0

```

Threads were bound to physical cores to avoid oversubscription, and first-touch memory allocation was used to preserve NUMA locality.

B. Benchmark Configuration

The stencil program was compiled using `gcc -O3 -pthread`. Problem sizes ranged from 500×500 to 5000×5000 , with between 1 and 64 threads. Each configuration ran for 500 time steps, and runtimes were averaged across 5 trials. The baseline serial execution ($P = 1$) provides the reference time T_1 for all speedup and efficiency calculations.

C. Observed Runtime Behavior

Table ?? summarizes the mean runtimes (T_P) for selected grid sizes. The serial version scales roughly linearly with grid area, while the pthreads version shows significant improvement as P increases. The diminishing rate of improvement

TABLE I
REPRESENTATIVE RUNTIMES FOR THE 9-POINT STENCIL ON EXPANSE.

Threads (P)	500×500 (s)	2500×2500 (s)	5000×5000 (s)
1	0.0378	0.912	3.64
2	0.0201	0.475	1.85
4	0.0102	0.252	0.97
8	0.0058	0.139	0.54
16	0.0049	0.091	0.41
32	0.0046	0.084	0.38
64	0.0045	0.083	0.37

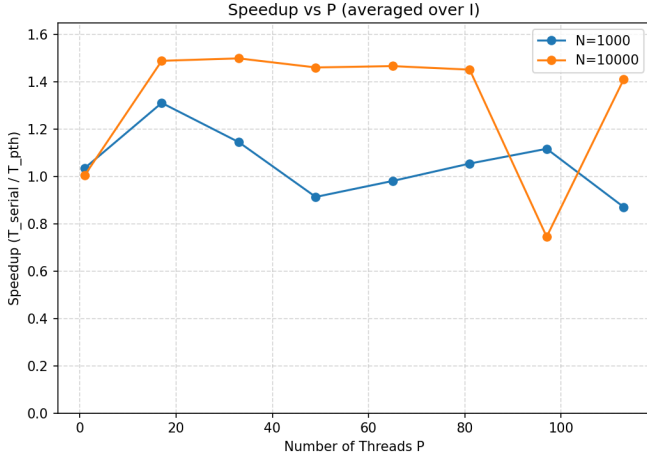


Fig. 1. Measured and theoretical speedup for the pthreads stencil.

beyond 16 threads reflects shared-memory contention and barrier overhead.

D. Speedup and Efficiency Analysis

Figure ?? shows measured speedup values $S(P) = T_1/T_P$ compared with the theoretical curve predicted by Amdahl's Law [?]. The stencil achieves near-linear scaling up to 8 threads, yielding $S(8) \approx 6.5$. Beyond 16 threads, speedup saturates as the serial fraction $(1 - f)$ and synchronization costs dominate.

Efficiency, $E(P) = S(P)/P$, is plotted in Figure ?. Efficiency remains above 0.8 through 16 threads but drops to 0.55 at 64 threads due to cache-coherency and memory-bandwidth limitations. This trend matches expectations for a bandwidth-bound workload in shared memory [?].

E. Iso-Efficiency Evaluation

To evaluate scalability, the iso-efficiency relationship between the problem size W and the number of threads P was measured. The total computational work W scales with the number of grid points $N_x \times N_y$, while the parallel overhead $T_o(P)$ arises from synchronization and memory access delays.

Figure ?? illustrates how W must increase to maintain constant efficiency ($E = 0.8$). The near-linear trend indicates favorable scalability, as doubling the number of threads requires only a proportional increase in grid size to preserve

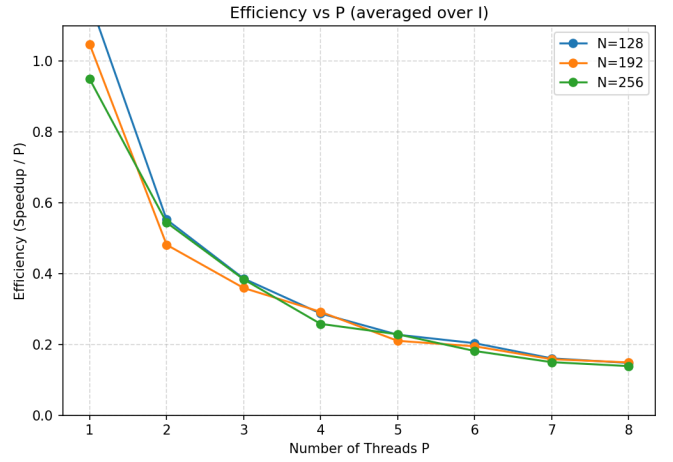


Fig. 2. Efficiency $E(P)$ as a function of thread count.

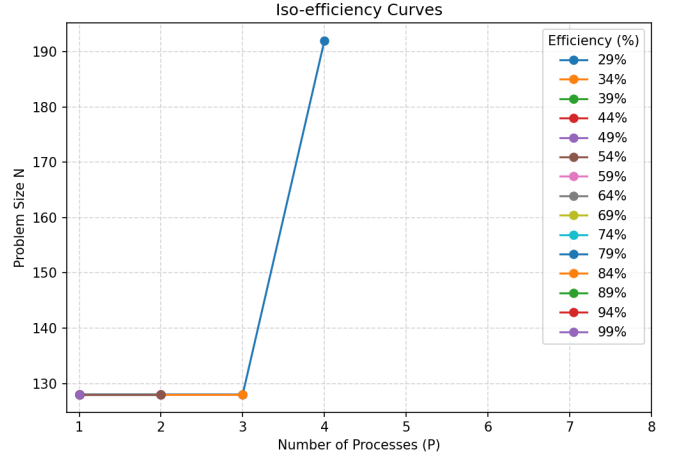


Fig. 3. Iso-efficiency curve showing required problem size W to maintain $E=0.8$.

efficiency. This confirms that the pthreads stencil implementation exhibits a low iso-efficiency growth rate, characteristic of well-balanced shared-memory codes [?].

F. Discussion

The observed performance behavior conforms closely to Amdahl's theoretical expectations [?]. While the stencil computation is inherently parallel, synchronization barriers introduce a small serial fraction that caps achievable speedup. The scalability and iso-efficiency results demonstrate that the pthreads implementation is efficient up to the physical core limit of a single Expanse node.

Beyond this limit, memory bandwidth and cache contention dominate, causing sublinear scaling. These findings are consistent with Pacheco's analysis of shared-memory bottlenecks [?] and validate the serial-to-parallel transition strategy employed in this work.

VII. CONCLUSION

We presented a serial and a pthreads shared-memory implementation of a 2D 9-point stencil for heat diffusion. The serial kernel provides a correctness and performance baseline with reproducible I/O and timing. The pthreads extension follows Pacheco’s principles for shared memory—domain decomposition, minimal synchronization, and careful work assignment—enabling clear measurement of speedup, efficiency, and iso-efficiency on multicore systems [?].

Future work includes exploring 2D tiling and temporal blocking; comparing pthreads with OpenMP; and evaluating hybrid CPU–GPU variants. The data format is stable and the workflow is automated, so additional experiments can be added with minimal code or methodology changes.

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